



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (general handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No : BIO F212
Course Title : Microbiology
Instructor-In charge : PRABHAT NATH JHA (Email:prabhatjha@pilani.bits-pilani.ac.in; Room: 3277-V)
Team of Instructors : Jitendra Panwar, Mohit Rana, Subhojit Roy, Swati Sorab Manjil, Anun Pandit, Nidhi Hallikeri

1. Course Description:

Introduction and classification of microbes; isolation, cultivation, physiological and biochemical characterization of microbes; structure, physiology and genetics of microbial cell; synthetic microbiology; host parasite relationship; physical chemical methods of controlling microbes; antimicrobial drugs; clinical microbiology; microbiology of soil, water and food; and related lab components.

2. Scope & Objective:

This course deals with the structure, physiology, genetics and growth of various microorganisms as well as their control. Emphasis will be given on microbes and their role in human health, environment and industry.

3. Text Book:

Tortora, G.J., Funke, B.R. and Case, C.L. 2020. Microbiology: An Introduction, 12th Ed., Pearson India Education Services Pvt. Ltd. India

4. Reference books:

Willey, J.M., Sherwood, L.M. and Woolverton, C.J. 2017. Prescott, Harley and Klein's Microbiology, 7th Edition, The McGraw-Hill Companies Inc., New York.

5. Lab Manual:

Experimental write-ups will be provided.

6. Course Plan

Module No.	Lecture/ Tutorial Session	Reference	Learning Outcome
1. Course layout and introduction to microbiology	L 1-3. The microbial world	TB-1, RB-1	Understanding the role of microbes in our life
2. Methods in Microbiology	L 4. Microscopy and Specimen preparation	TB-3, RB-2	Knowledge about microscopy & preparing microbial samples for observation
	L 5-6. Requirement for growth, obtaining pure cultures and maintenance	TB-6, RB-7	Knowledge about various culture media and methods to obtain pure microbial cultures
3. Study of Microbial Structures	L 7-10. The morphology & fine structure of bacteria	TB-4, RB-3	Understanding size, shape & arrangement of bacterial cells and their structures
	L 11-13. Eukaryotic microorganisms	TB-12, RB-4	Understanding fungi, algae and lichens



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4. Identifying Microorganisms	L 14-16. Characterization, classification and identification of microorganism	TB-10, RB-19	Understanding methods of microbial classification & identification
5. Virology	L 17-19. Virus, Viroids, Prions	TB-13, RB-6,27	Understanding features, type, taxonomy and replication of virus
6. Microbial Growth	L 20-22. Growth of Microbes and its measurement; Synchronous and continuous culture	TB-6, RB-7	Understanding bacterial growth curve, methods of microbial growth measurement
7. Microbial Physiology	L 23-25. Microbial metabolism	TB-5, RB-10, 11, 12	Knowledge of energy generation and utilization in microbes
8. Control of Microorganism	L 26-28. Physical and chemical methods of microbial control, Antimicrobial drugs	TB-7, 20 RB-8	Knowing the methods to prevent microbial growth
9. Microbial Genetics	L 29-32. The genetics of microorganisms; Synthetic Microbiology	TB-8, RB-16	Understanding genetic transformation & recombination in microbes and approaches in synthetic microbiology for applied purposes.
10. Clinical Microbiology	L 33-35. Principles of diseases and epidemiology, Microbial Mechanisms of Pathogenicity	TB-14, 15	Knowledge about bacterial pathogenesis in humans
11. Environmental Microbiology	L 36-38. Microbiology of soil, domestic and waste water	TB-27 RB-43	Understanding the role & application of microbes in environment
12. Applied Microbiology	L 39-41. Microbiology of food and Industrial microbiology	TB-28 RB-42	Understanding applications of microbes in food & beverage industry

7. Lab Components:

- Exp. 1. Introduction to microbiology laboratory and practices
- Exp. 2: Media preparation, Sterilization, Plating
- Exp. 3: Isolation of pure cultures of bacteria from various samples and cell count
- Exp. 4: Gram's staining of bacteria
- Exp. 5: Staining of fungus- Lactophenol cotton blue staining
- Exp. 6: Dehydrogenase activity assay for qualitative determination of microbial population
- Exp. 7: IMViC test/Carbohydrate Utilization test for biochemical characterization of bacteria
- Exp. 8: Bacterial growth curve, Effect of pH, salt on the growth of microorganisms
- Exp. 9: Enumeration of microbial cells in Neubauer chamber
- Exp. 10: Enumeration of bacteriophage (Phage titration kit) or Isolation of bacteriophages
- Exp. 11: Test of antimicrobial activity (zone inhibition, MIC)
- Exp. 12: Coliform counts in contaminated water sample

Note: Out of the above-mentioned list, a minimum of 10 experiments will be conducted in the Semester as per the availability of the consumables.



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8. Evaluation Scheme:

S. No.	Evaluation Component*	Weightage (%)	Date & Time	Remarks (Close Book/ Open Book)
1.	Mid-Semester Test	20	08/10/ FN1	CB
2.	Quizzes	10		CB
3.	Assignments	5		OB
4.	Laboratory evaluation	30		
5.	Comprehensive	35	08/12 FN	CB/OB

* Material in soft-copy will not be allowed in any evaluation component. Only your own handwritten notes will be allowed in OB exam; printouts/xerox of lecture slides and class notes will NOT be allowed.

Chamber consultation hour: To be announced in the class.

Notices: All notices will be sent to your BITS email address.

Make-up policy: If a student misses the Mid-Semester or Comprehensive Examination due to a genuine reason (serious medical causes leading to hospitalization (certified by doctor from BITS medical center), personal/family emergencies) there exists a provision to apply for make-up. Prior permission must be taken from the Instructor-in-Charge, before applying for make-up. Make-up decisions will be made on a case-by-case basis and only genuine cases as determined by the Departmental team will be considered. No Make-Up will be granted for quiz, assignments and Lab components.

Grading Policy: Award of grades would be guided by the histogram of marks and course average. "If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student will not be awarded a valid grade and the case will be reported as "NC"."

Note: It shall be the responsibility of the individual student to be regular in attending lectures and lab components as per the schedule announced.

Instructor-in-charge
BIO F212